

AMER ABU-ISSA

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CAREER OBJECTIVE

Seeking a graduate field engineering role in the electric and hybrid vehicle sector, applying formal mechanical and mechatronic engineering training to high-voltage diagnostic, calibration, and commissioning work. I offer an engineering organisation a graduate who moves fluidly between design-stage analysis and safety-critical, in-field electrical and mechanical execution from day one, built on hands-on experience with CAN bus diagnostics, ECU programming, and hybrid/high-voltage powertrain systems from an automotive diagnostics placement, paired with an ongoing electrical trade qualification in AS/NZS 3000 compliance and current safety-critical hardware experience.

EDUCATION

Bachelor of Engineering (Honours), Mechanical & Mechatronic Engineering *Mar 2020 – Dec 2026 (expected)*

University of Technology Sydney

Diploma in Professional Engineering Practice | Distinction average

Electrical Trade Units (concurrent) *2025 – Present*

TAFE NSW

AS/NZS 3000 wiring compliance, circuit and switchboard design, load calculations, earthing and protection systems, WHS documentation.

PROFESSIONAL EXPERIENCE

Electronic Monitoring Technician *Jan 2026 – Present*

Buddi, Rhodes NSW

- Configure and program dual-SIM GPS monitoring devices for accurate location tracking and correct alert generation, including exclusion-zone breaches and tamper detection.
- Maintain a zero-tolerance standard for missed or incorrect alerts in a safety-critical monitoring environment.

Automotive Diagnostics Intern (WIL 2) *2025*

Neysie Performance

- Diagnosed CAN bus faults, programmed ECUs, and calibrated ADAS systems on live customer vehicles.
- Worked on hybrid and high-voltage powertrain systems alongside mechanical tasks including engine disassembly and timing chain replacement.
- Led diagnostic project work and supported technician mentoring as part of structured SMART goals mapped to Engineers Australia Stage 2 competencies.

Assistant Manager *Jan 2021 – 2025*

EzyMart Convenience, Olympic Park

- Managed rostering, stock control, and training of new staff while studying full-time.

Apprentice Mechanic *Feb 2020 – Jun 2020*

All Parts Auto Pty Ltd, Smithfield

- Serviced vehicles and built foundational knowledge of automotive components and diagnostic tools.

KEY PROJECTS

- **SafeCo DishBot (UTS 41013):** Designed and programmed a pick-and-place robot workcell for a dish-washing scullery scenario using a UR3e, as part of a 3-person team.
- **500 kV Double-Circuit Transmission Line Design (UTS 48572):** Group report covering three-phase system analysis, transformer impedance testing, and R-L-C circuit modelling.
- **Solar Parabolic BBQ (Mechanical Systems Design Studio):** Designed a portable solar BBQ system through the ADAP³D design process.
- **12V-40V Boost Converter (Power Electronics):** Designed, built, and documented a MOSFET-based boost converter, including circuit schematics and switching analysis.

TECHNICAL SKILLS

CAN bus diagnostics, ECU programming, oscilloscope analysis, ADAS calibration, hybrid & high-voltage powertrain systems, AS/NZS 3000 electrical compliance, circuit design & load calculations, SolidWorks CAD & FEA, Python/C++ (ROS2), embedded hardware configuration (GPS/GSM)

CERTIFICATIONS

SafeWork NSW: National WHS General Construction Induction (CPCCWHS1001), April 2021

TAFE NSW Electrical Units: UEECD0051SA, UEEEL0021SA, UEEEL0021MC, UEEEL0003PR8 (in progress)